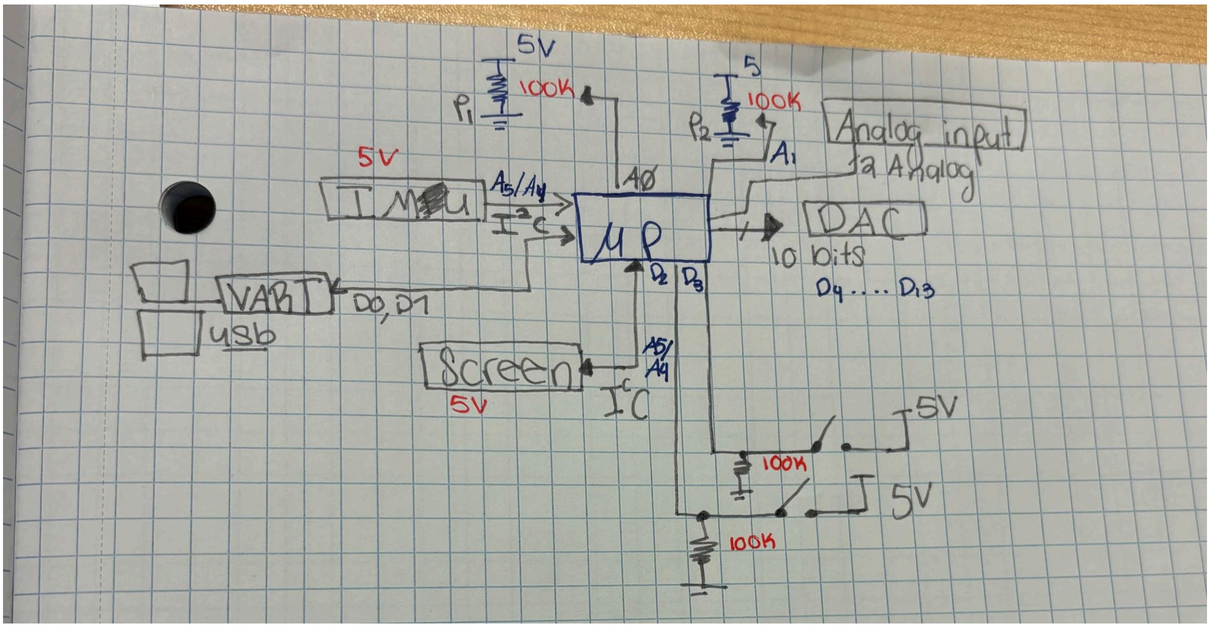
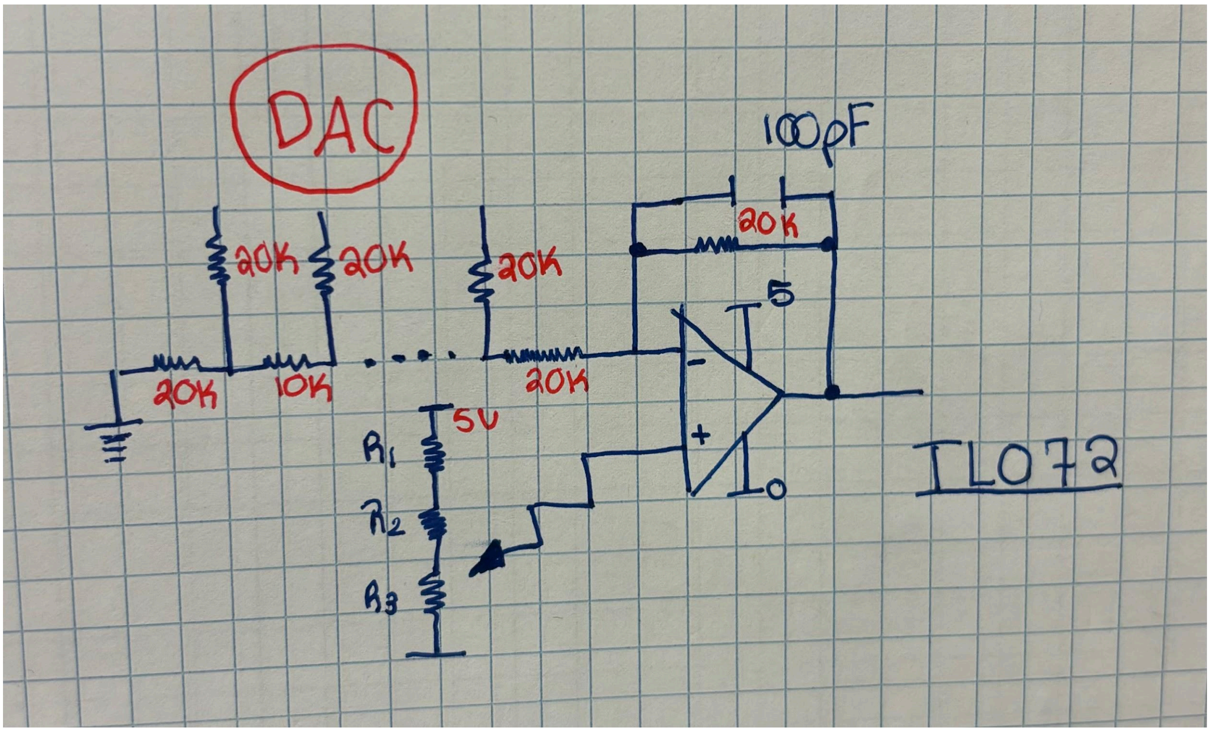
Introduction

The purpose of this project is to get familiar with the design of an Arduino Uno shield. Through this process,we can apply key embedded systems concepts, including interrupt routines, timers, mixed-signal processing (ADCs and DACs), and digital communication protocols such as I2C, SPI, UART, and CAN.

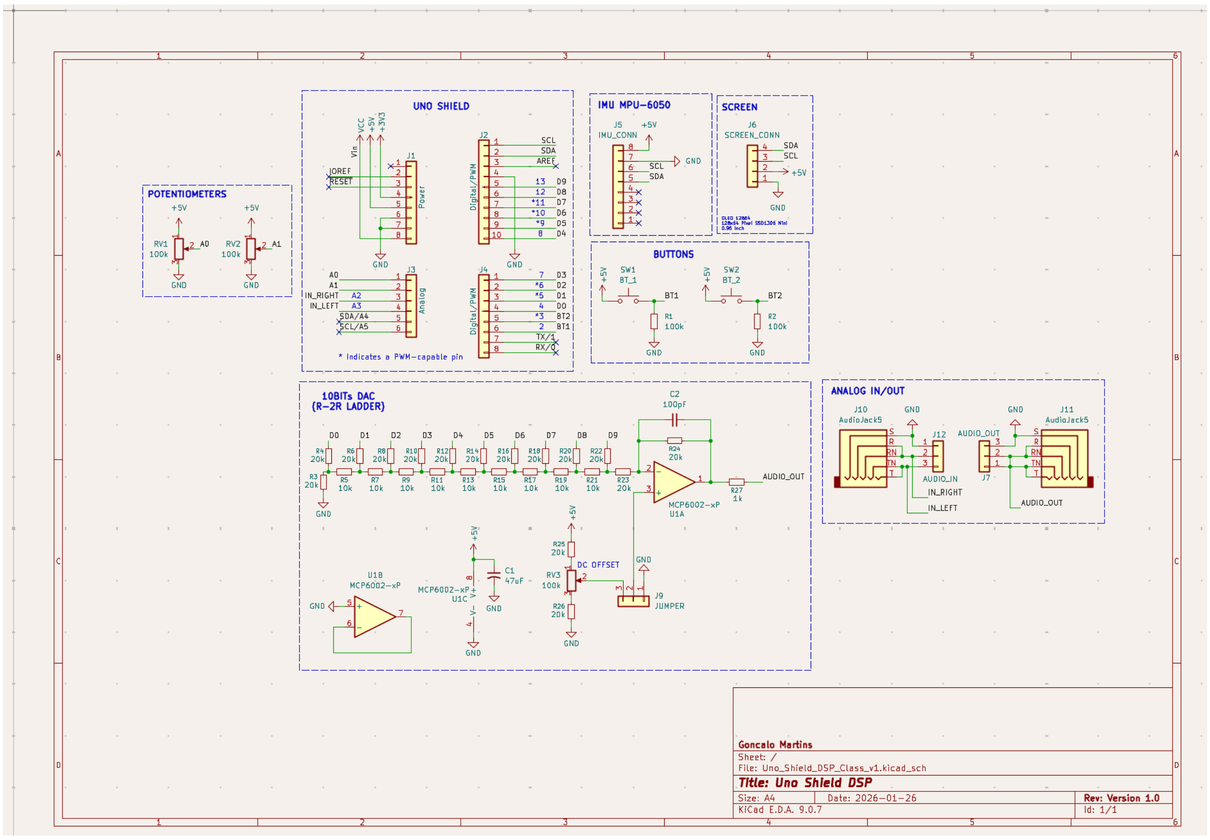
General block diagram:



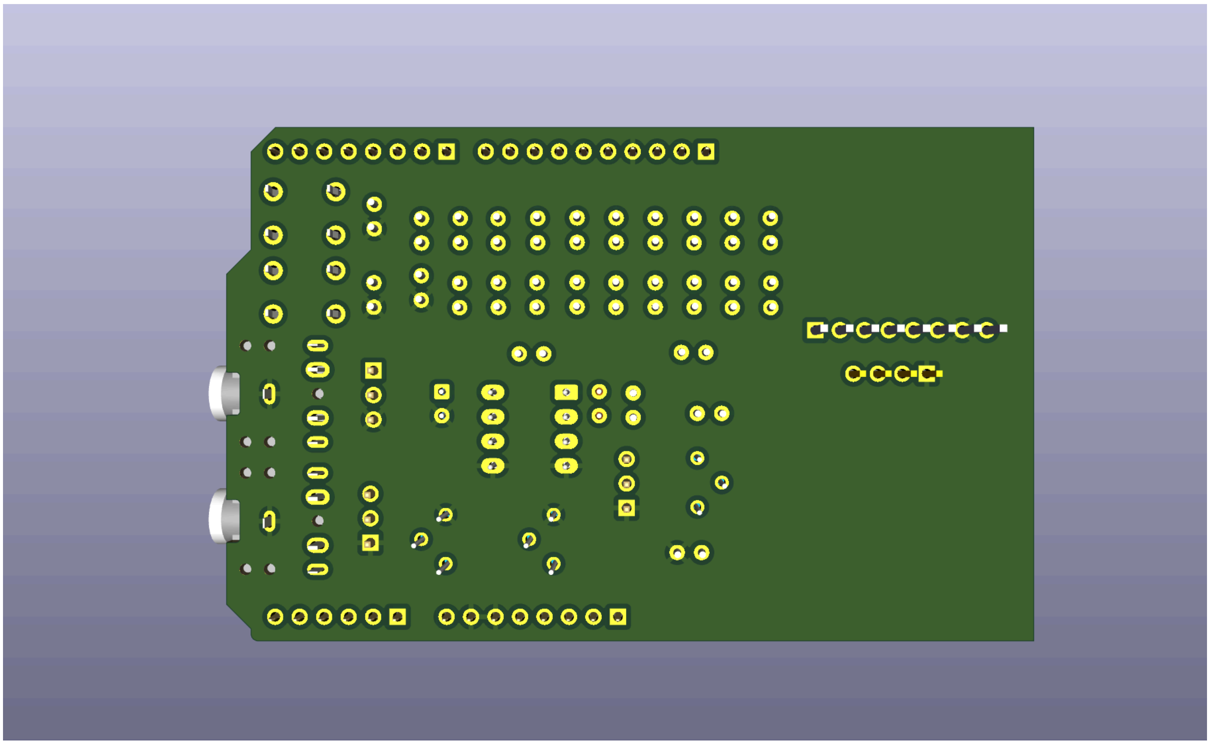
10bits R-2R Ladder DAC preliminary schematic:

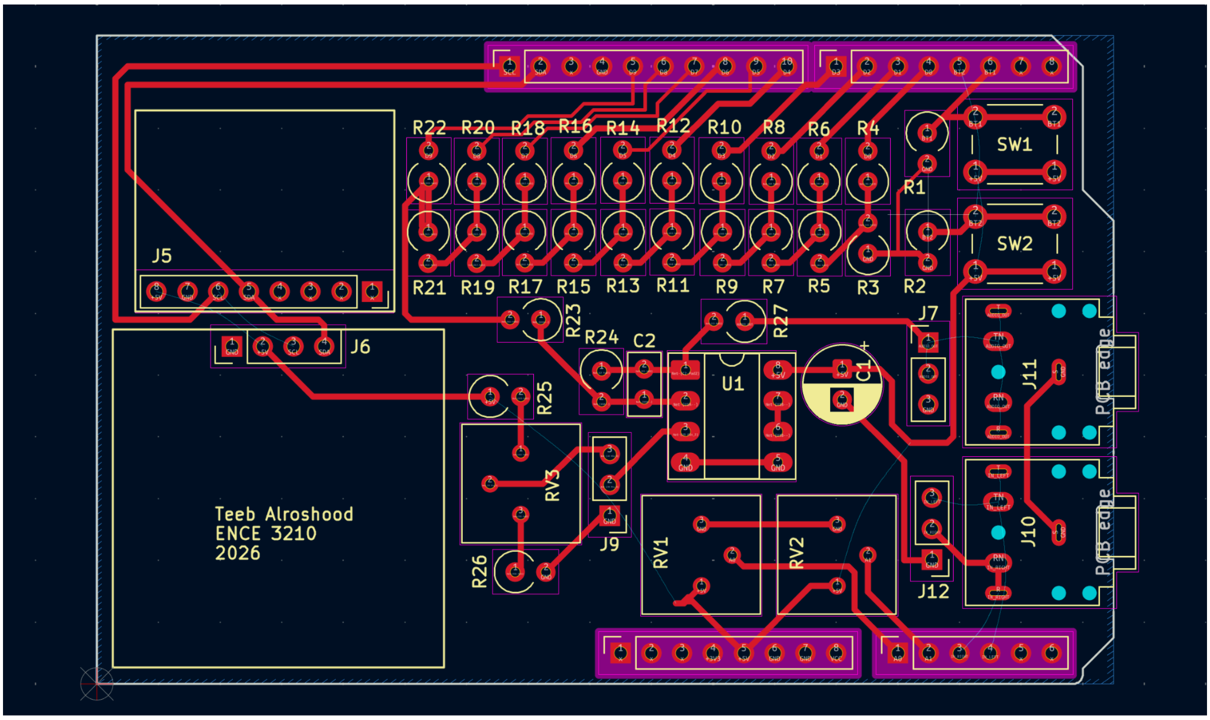


PCB Design (schematics)

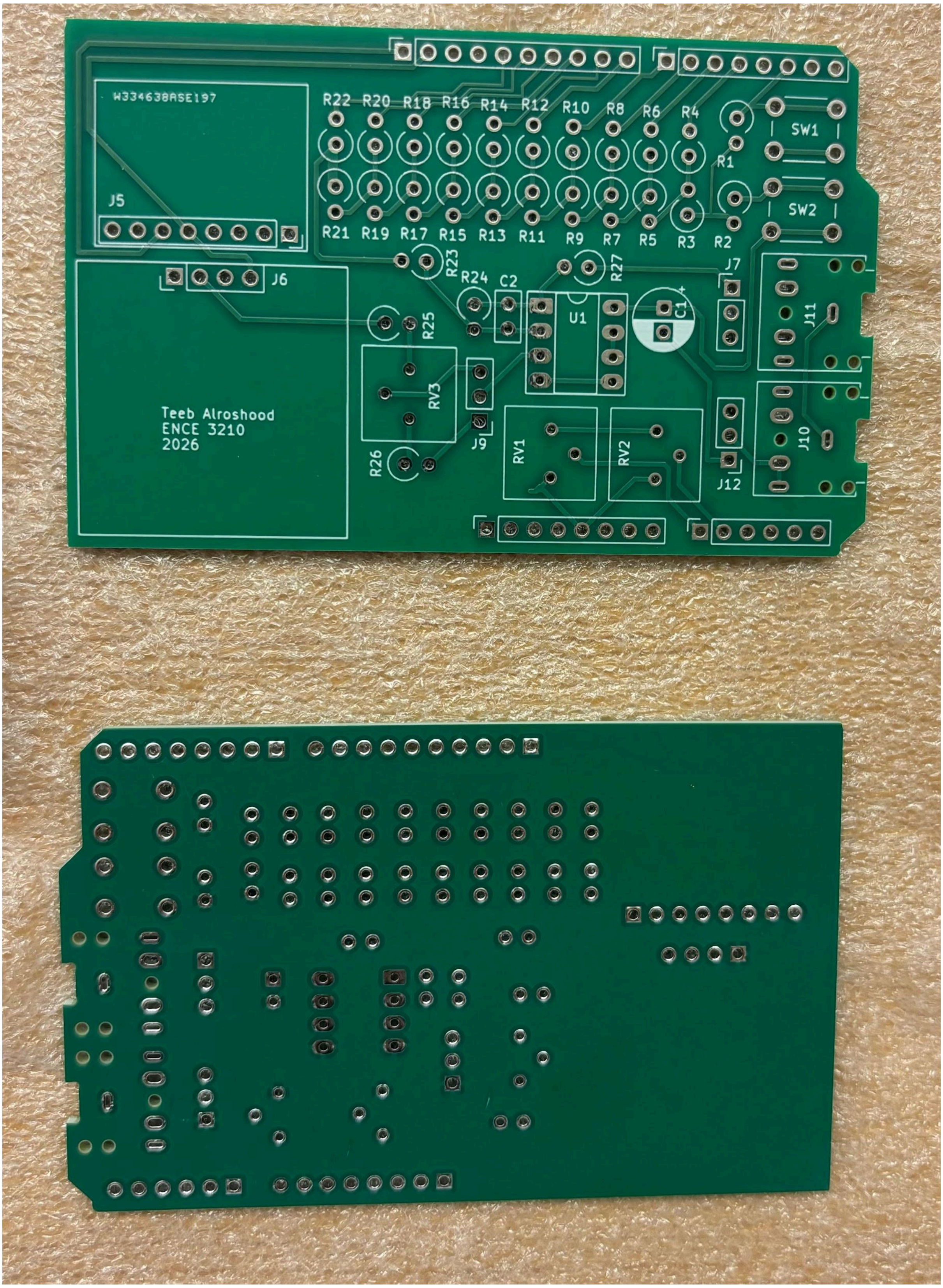


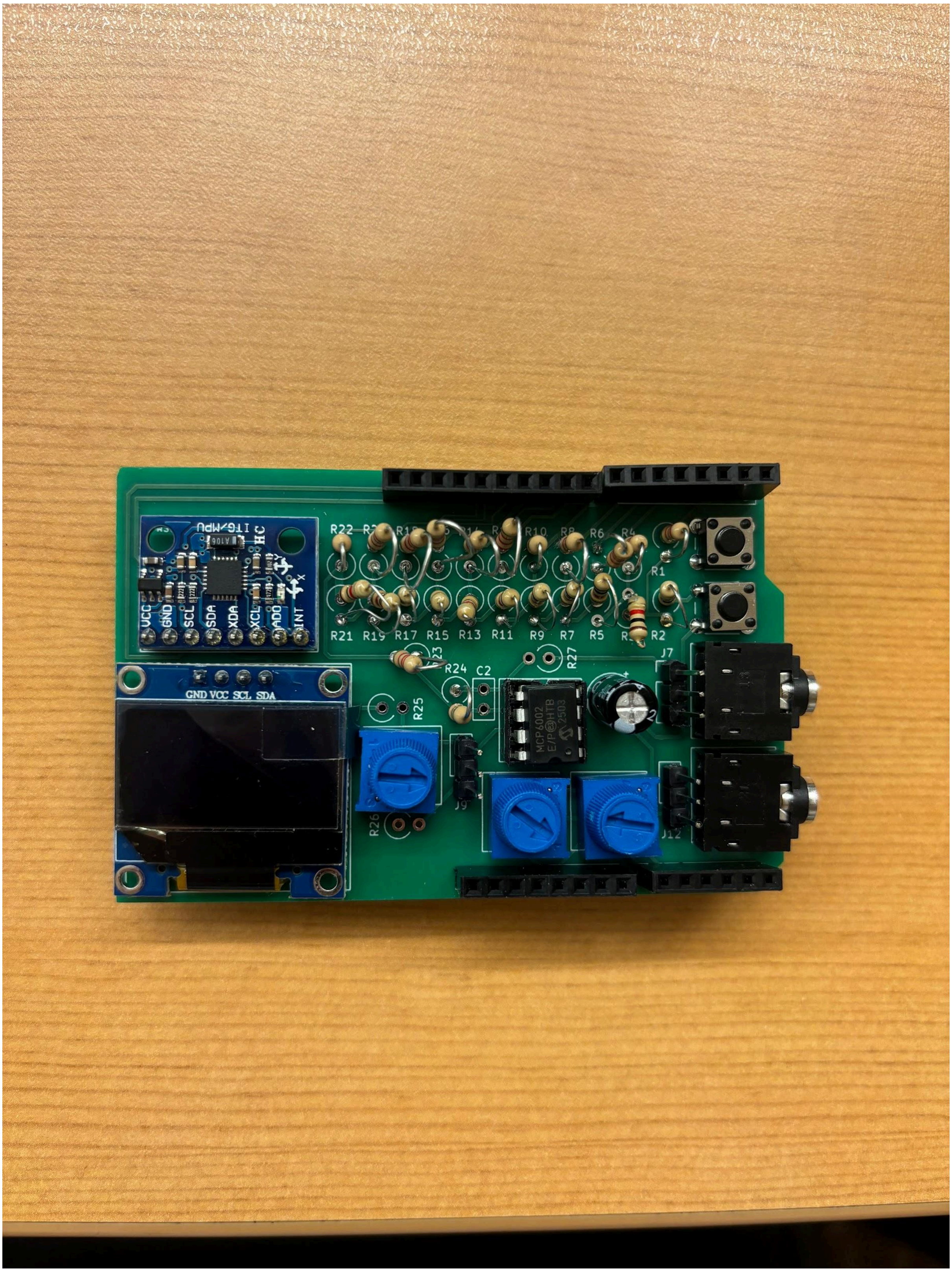
Layout:





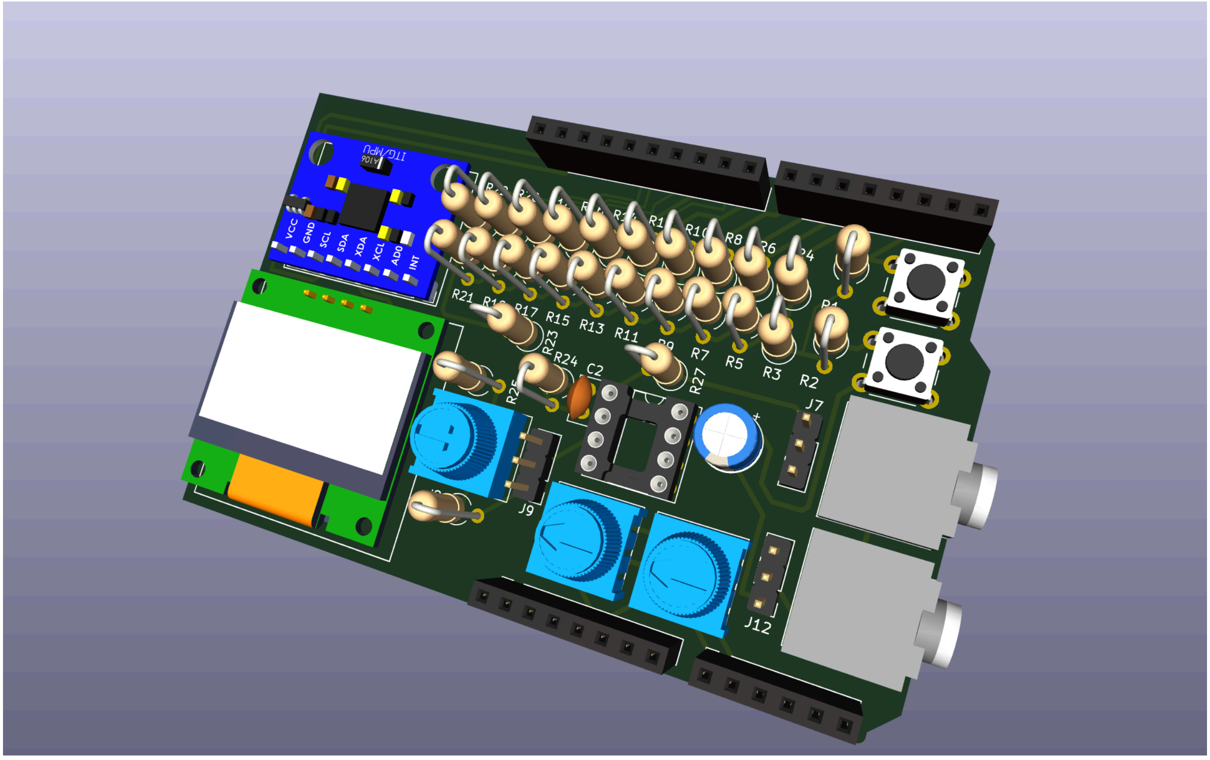
Assembly: These boards were assembled by PCBWay


W334638ASE197
R22 R20 R18 R16 R14 R12 R10 R8 R6 R4
R1
R21 R19 R17 R15 R13 R11 R9 R7 R5 R3 R2
SW1
SW2
J5
J6
R23
R24
C2
R27
J7
U1
C1
R25
RV3
J9
J11
J10
RV1
RV2
J12
R26
Teeb Alroshood
ENCE 3210
2026



Some modifications were made to the circuit board because it had been incorrectly soldered, which resulted in a short circuit. The board was adjusted and corrected to resolve this issue and restore proper functionality.

3D View:


ITG/MPU
VCC
GND
SCL
SDA
XDA
XCL
AD0
INT
R21
R17
R15
R13
R11
R9
R7
R5
R3
R2
R10
R8
R6
R4
R23
R24
C2
R27
J7
J9
J12